

The Capacity Region of the Binary Symmetric Z-Interference Channel

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Abstract

We give the capacity region, in closed form, of the two-user binary symmetric Z-interference channel $Y_1 = X_1 \oplus X_2 \oplus N_1$, $Y_2 = X_2 \oplus N_2$ with independent noises $N_i \sim \text{Bern}(p_i)$, $p_1, p_2 \in (0, \frac{1}{2})$, for all parameters. For $p_1 \leq p_2$ the region is $\{R_1 + R_2 \leq 1 - h(p_1), R_2 \leq 1 - h(p_2)\}$, an instantiation of classical strong-interference results. For $p_1 > p_2$ (weak interference) the region equals that of the degraded binary symmetric broadcast channel with parameters (p_1, p_2) : $R_2 \leq 1 - h(p_2)$ and $R_1 \leq 1 - h(q \star h^{-1}(h(p_2) + R_2))$ with $q = (p_1 - p_2)/(1 - 2p_2)$; treating interference as noise with an optimized input bias is capacity-achieving and rate splitting is unnecessary. The weak-regime converse is an eight-line application of Mrs. Gerber's Lemma. The result evaluates, in closed form, the single-letter characterization implied by Liu–Goldsmith for this channel class.

1 Model and result

Two senders, two receivers: $Y_1 = X_1 \oplus X_2 \oplus N_1$, $Y_2 = X_2 \oplus N_2$, with $N_1 \sim \text{Bern}(p_1)$, $N_2 \sim \text{Bern}(p_2)$ independent of each other and of the inputs, $p_1, p_2 \in (0, \frac{1}{2})$; receiver i decodes message i only, and the encoders are independent. h is the binary entropy, $h^{-1} : [0, 1] \rightarrow [0, \frac{1}{2}]$ its inverse, $a \star b = a(1 - b) + b(1 - a)$.

Theorem 1. *The capacity region of the BS-ZIC is*

(i) if $p_1 \leq p_2$: $\{(R_1, R_2) \geq 0 : R_1 + R_2 \leq 1 - h(p_1), R_2 \leq 1 - h(p_2)\}$;

(ii) if $p_1 > p_2$, with $q = \frac{p_1 - p_2}{1 - 2p_2} \in (0, \frac{1}{2})$:

$$\{(R_1, R_2) \geq 0 : R_2 \leq 1 - h(p_2), R_1 \leq 1 - h(q \star h^{-1}(h(p_2) + R_2))\}.$$

The weak-regime sum capacity is $1 - h(p_2)$, attained only at $(0, 1 - h(p_2))$.

*Produced by AI agents (a Gemini-based agent proved the strong-interference regime; a Claude-based agent proved the weak regime and drafted this note; each regime was reviewed by another AI system) on <https://github.com/gokhanmergen/information-theory-problems>, overseen by the human maintainer. Code for all numerical checks is in the repository.

2 Proof of (i): strong-to-moderate interference

Converse. Since $p_1 \leq p_2$, write $p_2 = p_1 \star p_0$ and couple $N_2^n = N_1^n \oplus N_0^n$ with $N_0 \sim \text{Bern}(p_0)$ i.i.d. independent; this preserves the law of (X_2^n, Y_2^n) , so data processing gives $I(X_2^n; Y_2^n) \leq I(X_2^n; X_2^n \oplus N_1^n) = I(X_2^n; Y_1^n \mid X_1^n)$ (given X_1^n , Y_1^n is a bijection of $X_2^n \oplus N_1^n$; encoders independent). With Fano, $n(R_1 + R_2) \leq I(X_1^n, X_2^n; Y_1^n) + n\epsilon_n \leq n(1 - h(p_1)) + n\epsilon_n$, and $nR_2 \leq n(1 - h(p_2)) + n\epsilon_n$.

Achievability. Uniform inputs; receiver 1 jointly decodes both messages over the mod-2 MAC $(R_1, R_2, R_1 + R_2 \leq 1 - h(p_1))$; receiver 2 needs $R_2 \leq 1 - h(p_2)$, which implies the MAC's R_2 constraint when $p_1 \leq p_2$. Regions match [2, 3, 7]. $\square$

3 Proof of (ii): weak interference

Converse. Now $p_1 = p_2 \star q$; couple $N_1^n = N_2^n \oplus Q^n$, $Q \sim \text{Bern}(q)$ i.i.d. independent, preserving the law of (X_1^n, X_2^n, Y_1^n) . Let $V^n = X_2^n \oplus N_2^n$ (distributed as Y_2^n). Fano at receiver 2: $\frac{1}{n}H(V^n) \geq h(p_2) + R_2 - \epsilon_n$. Mrs. Gerber's Lemma [1]: $\frac{1}{n}H(V^n \oplus Q^n) \geq h(q \star h^{-1}(\frac{1}{n}H(V^n)))$. Hence

$$nR_1 \leq n - H(Y_1^n \mid X_1^n) + n\epsilon_n = n - H(V^n \oplus Q^n) + n\epsilon_n \leq n[1 - h(q \star h^{-1}(h(p_2) + R_2 - \epsilon_n))] + n\epsilon_n,$$

using $H(Y_1^n \mid X_1^n) = H(X_2^n \oplus N_1^n) = H(V^n \oplus Q^n)$ and monotonicity of $h(q \star h^{-1}(\cdot))$. Let $n \rightarrow \infty$.

Achievability. Treat interference as noise with a biased input: $X_2 \sim \text{Bern}(\pi)$ i.i.d., $X_1 \sim \text{Bern}(\frac{1}{2})$. Receiver 2: $R_2 = h(\pi \star p_2) - h(p_2)$. Receiver 1: $Y_1 = X_1 \oplus Z$ with $Z \sim \text{Bern}(\pi \star p_1)$, so $R_1 = 1 - h(\pi \star p_1)$. With $s = \pi \star p_2$ we have $h(s) = h(p_2) + R_2$ and, since $p_1 = p_2 \star q$, $\pi \star p_1 = s \star q$; hence $R_1 = 1 - h(q \star h^{-1}(h(p_2) + R_2))$ — the inner bound traces the converse curve exactly as π runs over $[0, \frac{1}{2}]$. Lower rates are obtained by using the same input law and coding below the two point-to-point mutual informations.

It remains to justify the sum-capacity statement in the theorem. Along the boundary,

$$R_1 + R_2 = 1 - h(p_2) + h(s) - h(q \star s).$$

For $s \in [p_2, \frac{1}{2})$ and $q > 0$, one has $s < q \star s < \frac{1}{2}$, hence $h(s) < h(q \star s)$. Thus the sum is strictly below $1 - h(p_2)$ except at $s = \frac{1}{2}$, which gives the unique corner $(0, 1 - h(p_2))$. $\square$

4 Discussion

In the weak regime the BS-ZIC region coincides with the degraded BSC broadcast region [1]: receiver 1 sees user 2's codeword exactly as a degraded broadcast "cloud," and Mrs. Gerber's Lemma converts the degradation into a converse, as in Wyner–Ziv. No rate splitting is needed anywhere in the parameter space, and weak-regime sum-optimality forces user 1 silent. The channel satisfies the class conditions of Liu–Goldsmith [4] exactly and in n -letter form (mod-2 additivity gives an invariant conditional output entropy, and the uniform input maximizes the interfered receiver's output entropy irrespective of the interference law), so part (ii) can be read as a closed-form evaluation of their characterization with a degenerate auxiliary. Moreover, after relabeling users and writing $N_1 \stackrel{d}{=} N_2 \oplus \text{Bern}(q)$, the weak regime is capacity-region equivalent to Benzel's discrete additive degraded interference channel [5]. Benzel's literal two-output additive law is different; the equivalence of the Z-channel form to that law is stated explicitly by Liu–Goldsmith and is consistent with the degraded-broadcast equivalence in [6]. Thus part (ii) should

be regarded as classical; the contribution of this note is the assembled closed-form statement for all parameters and the short self-contained Mrs. Gerber converse.

References

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